

Here are the different ways to create a Spring Boot project, from quickest to most manual:

1. Spring Initializr (Web UI)

- Go to start.spring.io
- Choose project metadata (Group, Artifact, Dependencies, Java version, Spring Boot version)
- Generate and download a ZIP file containing a pre-configured project structure
- Import into your IDE (IntelliJ, Eclipse, VS Code) as a Maven/Gradle project

Best for: Beginners or quick prototyping.

2. Spring Initializr via IDE Plugins

- IntelliJ IDEA Ultimate: Built-in Spring Initializr support → *New Project* → *Spring Initializr*
- Eclipse/STS: Install Spring Tools → *File* → *New* → *Spring Starter Project*
- VS Code: Use Spring Boot Extension Pack → *Spring Initializr* command palette option

Best for: Developers who prefer staying within their IDE.

3. Spring Boot CLI (Command Line Interface)

- Install Spring Boot CLI via SDKMAN (`sdk install springboot`) or Homebrew
- Create a project using:

```
bash
spring init --dependencies=web,data-jpa my-project
```
- Also allows writing Groovy scripts on the fly with `spring run app.groovy`

Best for: Scripting, quick experiments, or CI/CD automation.

4. Using Maven / Gradle from Scratch (Manual)

- Create a `pom.xml` (Maven) or `build.gradle` (Gradle) file
- Add `spring-boot-starter-parent` as parent (Maven) or `plugins` (Gradle)
- Define dependencies like `spring-boot-starter-web`

- Create the main application class
with `@SpringBootApplication` and `main()` method
- Run with `mvn spring-boot:run` or `gradle bootRun`

Best for: Learning internals or custom build setups.

5. Using Online Code Editors / Cloud IDEs

- Platforms like Gitpod, Coding Rooms, or Replit provide Spring Boot templates
- Some offer "one-click" starter projects with pre-installed dependencies

Best for: Collaborative coding or education environments.

6. Using Custom Scaffolding Tools

- Tools like JHipster or Micronaut Launch can generate Spring Boot projects with additional features (microservices, security, frontend)
- Though not pure Spring Initializr, they output a full Spring Boot application

Best for: Enterprise-grade, full-stack projects.

Recommendation:

Method	Speed	Flexibility
Spring Initializr (Web/IDE)	⚡ Fast	✅ High
CLI	⚡ Fast	✅ Medium
Manual Maven/Gradle	🐢 Slower	✅✅ Maximum

💡 Pro Tip: Always use Spring Initializr (web or IDE) for 95% of your projects — it gives you best practices, correct dependency versions, and a clean structure out of the box.